

Quality Assurance Test Summary Blueprint

E-Commerce Test Strategy & 135-Scenario Execution Matrix

Project Parameters

Project Name: Automation Exercise Capstone
Automation Stack: Playwright Node.js (JavaScript ES6+)
Website Target Link: <https://automationexercise.com/>

Project Objectives

- **Achieve Targeted Coverage Goals:** Outline, map, and baseline 120 traceable test conditions to eliminate functional script blind spots across the target system.
- **Standardize Automation Code Architecture:** Structurally organize testing parameters to leverage asynchronous worker isolation, parallel execution threads, and clean browser context state controls.
- **Validate Transactional Math Calculations:** Ensure complete calculation precision across shopping cart matrices, calculated row totals, and localized checkout balances.
- **Assess Form Fields Resilience:** Validate input field boundaries, string thresholds, type restrictions, and data sanitization layers against unauthorized injection behavior.

Detailed Project Flow

Step 1. Environment Launch: Initialize Playwright browser instances and navigate to the base URL environment across isolated asynchronous workers.

Step 2. Identity Management: Execute new user registration with profile details or authenticate existing user credentials via the secure Login portal.

Step 3. Catalog Interrogation: Navigate categories, utilize search string queries to locate targeted items, and perform item layout validations.

Step 4. Basket Consolidation: Add items to the shopping cart, update real-time unit quantities, and verify multi-product mathematical alignments.

Step 5. Checkout Processing: Advance past the basket view, cross-verify profile shipping address maps, enter description text, and process payment card info.

Step 6. Document Validation & Exit: Launch the text invoice download utility, scrape the local file to verify calculations match, and safely trigger logout actions.

Step 7. Lifecycle Profile Erasure: Authenticate session access one final time and execute explicit account deletion workflows to verify complete profile isolation.

Test Cases & Criticality

Functionality	Key Test (What We Are Going To Do)	Test Cases	Criticality
User Onboarding	Verify registration data handoffs, empty mandatory text parameters, field highlights, and script tag escaping.	25	CRITICAL
Account Security	Validate logins, case-sensitivity bounds, URL parameters parameter block jumps, and brute force throttling.	20	CRITICAL
Product Catalog	Assert item name text fields, check currency formats, and test quantitative input boundaries up to 9999 units.	15	HIGH
Cart Management	Track line accumulations, multi-row matrix calculations, session reloads, and empty cart placeholder elements.	15	HIGH
Checkout Flow	Check character-for-character shipping vs billing maps and run maximum 1000 character block comment tests.	15	HIGH
Invoicing Systems	Verify secure month/year expiration masks, track HTTP 200 return headers, and parse text file receipts.	18	CRITICAL
Contact us	Validate basic communication forms, test multi-click actions, and drop unsafe executable file paths safely.	12	MEDIUM
Account Lifecycle	Trigger removals after checkout, deny access using recycled profiles, and force resets to zero items.	15	HIGH

Final Project Deliverables

- **Master Test Plan & Strategy Document:** Comprehensive specification blueprint containing the precise scope of work, testing schedules, and structural case definitions.
- **Automated Playwright JS Test Suite:** Modular, maintainable JavaScript source code tracking precisely against the defined 135 testing objectives using the POM design pattern.
- **Data-Driven JSON Injection Arrays:** External data configuration files populated with edge-case criteria, formatting boundary strings, and sanitization inputs.
- **Automated Test Execution Reporting Portals:** Localized HTML dashboards and reports detailing test logs, diagnostic screenshots, and stack traces.

- **Invoice Content Data Scrapping Parser Utility:** A robust file automation helper script that extracts and cross-checks payment values inside downloaded text assets.